

## SAFETY DATA SHEET

Revision Date 30-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** Nickel fluoride tetrahydrate

**Cat No. :** 14132

**CAS No** 13940-83-5  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                      |             |
|------------------------------------------------------|-------------|
| Respiratory Sensitization                            | Category 1  |
| Skin Sensitization                                   | Category 1  |
| Germ Cell Mutagenicity                               | Category 2  |
| Carcinogenicity                                      | Category 1A |
| Reproductive Toxicity                                | Category 1A |
| Specific target organ toxicity - (repeated exposure) | Category 1  |
| Target Organs - Lungs.                               |             |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**

May cause an allergic skin reaction  
May cause allergy or asthma symptoms or breathing difficulties if inhaled  
Suspected of causing genetic defects  
May cause cancer  
May damage fertility or the unborn child  
Causes damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Use personal protective equipment as required  
In case of inadequate ventilation wear respiratory protection  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves  
Do not breathe dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product

**Response**

IF exposed or concerned: Get medical attention/advice

**Inhalation**

IF INHALED: If breathing is difficult, remove to fresh air and keep at rest in a position comfortable for breathing  
If experiencing respiratory symptoms: Call a POISON CENTER or doctor/physician

**Skin**

IF ON SKIN: Wash with plenty of soap and water  
If skin irritation or rash occurs: Get medical advice/attention  
Wash contaminated clothing before reuse

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Very toxic to aquatic life with long lasting effects  
WARNING. Cancer - <https://www.p65warnings.ca.gov/>.

### 3. Composition/Information on Ingredients

| Component                                         | CAS No     | Weight % |
|---------------------------------------------------|------------|----------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | 13940-83-5 | <=100    |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

|                                            |                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                                                                                       |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                                                                                                                                                                              |
| <b>Most important symptoms and effects</b> | May cause allergic skin reaction. May cause allergy or asthma symptoms or breathing difficulties if inhaled. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                              |

## 5. Fire-fighting measures

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Suitable Extinguishing Media</b>     | Not combustible.         |
| <b>Unsuitable Extinguishing Media</b>   | No information available |
| <b>Flash Point</b>                      | No information available |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

### Specific Hazards Arising from the Chemical

Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Hydrogen fluoride. Nickel oxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
-

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                                                          |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.                                                                                                                                                                                        |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.                                                                                                                                                                             |

## 7. Handling and storage

|                 |                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible                                                                                               |

Materials. Oxidizing agent.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component                                         | ACGIH TLV                                             | OSHA PEL                                                                     | NIOSH                                                                                  | Mexico OEL (TWA)                                      |
|---------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | TWA: 2.5 mg/m <sup>3</sup> TWA: 0.1 mg/m <sup>3</sup> | (Vacated) TWA: 0.1 mg/m <sup>3</sup><br>(Vacated) TWA: 2.5 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup> IDLH: 250 mg/m <sup>3</sup><br>TWA: 0.015 mg/m <sup>3</sup> | TWA: 2.5 mg/m <sup>3</sup> TWA: 0.1 mg/m <sup>3</sup> |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Ensure adequate ventilation, especially in confined areas.

### Personal Protective Equipment

#### Eye/face Protection

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

#### Respiratory Protection

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

#### Recommended Filter type:

Particulates filter conforming to EN 143.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                     |
|----------------------------------------|-------------------------------------|
| Physical State                         | Solid                               |
| Appearance                             | Green                               |
| Odor                                   | Odorless                            |
| Odor Threshold                         | No information available            |
| pH                                     | No information available            |
| Melting Point/Range                    | No data available                   |
| Boiling Point/Range                    | No information available            |
| Flash Point                            | No information available            |
| Evaporation Rate                       | Not applicable                      |
| Flammability (solid,gas)               | No information available            |
| Flammability or explosive limits       |                                     |
| Upper                                  | No data available                   |
| Lower                                  | No data available                   |
| Vapor Pressure                         | No information available            |
| Vapor Density                          | Not applicable                      |
| Specific Gravity                       | 4.6 g/cm <sup>3</sup>               |
| Solubility                             | No information available            |
| Partition coefficient; n-octanol/water | No data available                   |
| Autoignition Temperature               | No information available            |
| Decomposition Temperature              | No information available            |
| Viscosity                              | Not applicable                      |
| Molecular Formula                      | NiF <sub>2</sub> .4H <sub>2</sub> O |
| Molecular Weight                       | 168.77 (96.71anhy)                  |

## 10. Stability and reactivity

|                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available |
| <b>Stability</b>                        | Stable under normal conditions.            |
| <b>Conditions to Avoid</b>              | Incompatible products.                     |
| <b>Incompatible Materials</b>           | Oxidizing agent                            |
| <b>Hazardous Decomposition Products</b> | Hydrogen fluoride, Nickel oxides           |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.   |
| <b>Hazardous Reactions</b>              | None under normal processing.              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | No information available                                                                 |
| <b>Sensitization</b>   | May cause sensitization by skin contact                                                  |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                         | CAS No     | IARC       | NTP   | ACGIH      | OSHA       | Mexico     |
|---------------------------------------------------|------------|------------|-------|------------|------------|------------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | 13940-83-5 | Not listed | Known | Not listed | Not listed | Not listed |

*NTP: (National Toxicity Program)*

*NTP: (National Toxicity Program)*

*Known - Known Carcinogen*

*Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen*

|                                                   |                                                                                                                                                                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mutagenic Effects</b>                          | No information available                                                                                                                                                              |
| <b>Reproductive Effects</b>                       | No information available.                                                                                                                                                             |
| <b>Developmental Effects</b>                      | No information available.                                                                                                                                                             |
| <b>Teratogenicity</b>                             | No information available.                                                                                                                                                             |
| <b>STOT - single exposure</b>                     | None known                                                                                                                                                                            |
| <b>STOT - repeated exposure</b>                   | Lungs                                                                                                                                                                                 |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                              |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                              |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                        |

## 12. Ecological information

**Ecotoxicity**

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

**Persistence and Degradability** based on information available. May persist

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

**DOT**

**UN-No** UN3288  
**Proper Shipping Name** Toxic solid, inorganic, n.o.s.  
**Technical Name** (Nickel(II) fluoride tetrahydrate)  
**Hazard Class** 6.1  
**Packing Group** III

**TDG**

**UN-No** UN3288  
**Proper Shipping Name** Toxic solid, inorganic, n.o.s.  
**Hazard Class** 6.1  
**Packing Group** III

**IATA**

**UN-No** UN3288  
**Proper Shipping Name** Toxic solid, inorganic, n.o.s.  
**Hazard Class** 6.1  
**Packing Group** III

**IMDG/IMO**

**UN-No** UN3288  
**Proper Shipping Name** Toxic solid, inorganic, n.o.s.  
**Hazard Class** 6.1  
**Packing Group** III

### 15. Regulatory information

**United States of America Inventory**

| Component                                         | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|---------------------------------------------------|------------|------|-----------------------------------------------|-----------------------------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | 13940-83-5 | -    | -                                             | -                           |

**Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea

(KECL).

| Component                                         | CAS No     | DSL | NDSL | EINECS | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|---------------------------------------------------|------------|-----|------|--------|-------|------|------|------|-------|------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | 13940-83-5 | -   | -    | -      | -     | -    |      | X    | X     | -    |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component                                         | CAS No     | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|---------------------------------------------------|------------|----------|-------------------------------|---------------------------------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | 13940-83-5 | <=100    | 0.1 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

| Component                                         | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|---------------------------------------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | -                          | -                           | X                      | -                         |

**Clean Air Act**

| Component                                         | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|---------------------------------------------------|-----------|-------------------------|-------------------------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | X         |                         | -                       |

OSHA - Occupational Safety and Health Administration

Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65**

This product contains the following Proposition 65 chemicals.

| Component                                         | CAS No     | California Prop. 65                              | Prop 65 NSRL | Category                    |
|---------------------------------------------------|------------|--------------------------------------------------|--------------|-----------------------------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | 13940-83-5 | Carcinogen<br>Developmental<br>Male Reproductive | -            | Developmental<br>Carcinogen |

**U.S. State Right-to-Know Regulations**

| Component                                         | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|---------------------------------------------------|---------------|------------|--------------|----------|--------------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | -             | X          | X            | X        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations****Mexico - Grade**

No information available

**Authorisation/Restrictions according to EU REACH**

| Component                                         | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------------------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | 13940-83-5 | -                                                                   | Use restricted. See item 27. (see link for restriction details)               | -                                                                                                     |

**REACH links**<https://echa.europa.eu/substances-restricted-under-reach>**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                         | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | 13940-83-5 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component                                         | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Nickel fluoride (NiF <sub>2</sub> ), tetrahydrate | 13940-83-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information****Prepared By**Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com**Revision Date**

30-Mar-2024

**Print Date**

30-Mar-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other

materials or in any process, unless specified in the text

**End of SDS**